GLM Logistic Regression of Highly Rated Amazon Electronic Products

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Overview

Audience: An Amazon Marketing Executive, particularly in the Electronics Division. They want to see insights about how to best market the products to lead to highly-rated reviews, because high reviews increase sales for that product. We hope to give them recommendations for specific actions they can take from a marketing perspective to increase the odds of a review 4.5/5 stars and above.

Data: 2025 Amazon Electronic Product Sales Kaggle Dataset.

- amazon_products_sales_data_uncleaned.csv file.

Cleaning Dataset

	rating	number_of_reviews	current/discounted_price	listed_price	is_best_seller	is_sponsored	is_couponed	buy_box_availability
1	4.6 out of 5 stars	375	89.68	$159.00	No Badge	Sponsored	Save 15% with coupon	Add to cart
2	4.3 out of 5 stars	2,457	9.99	$15.99	No Badge	Sponsored	No Coupon	Add to cart
3	4.6 out of 5 stars	3,044	314.00	$349.00	No Badge	Sponsored	No Coupon	Add to cart
4	4.6 out of 5 stars	35,882		No Discount	Best Seller	Organic	No Coupon	
5	4.8 out of 5 stars	28,988		No Discount	No Badge	Organic	No Coupon	

	is_sponsored	clean_number_reviews	is_discount	clean_current_discounted_price	is_best_seller_clean	rating_cat	is_couponed_clean	buy_box_clean
1	Sponsored	375	Discount	89.68	No	1	Yes	Yes
2	Sponsored	2457	Discount	9.99	No	0	No	Yes
3	Sponsored	3044	Discount	314	No	1	No	Yes
4	Organic	4380	No Discount	16.99	No	0	Yes	No
5	Organic	865598	No Discount	14.99	Best Seller	1	No	Yes

Clean Dataset Description

- **Size** is 29,289x8

X variables

- *number_reviews* – Total number of customer reviews (N)
- *current_discounted_price* – Current price after discount (N)
- *is_discount* – Listed price after discount (B)
- *Is_best_seller* – Indicates if the product is tagged as a Best Seller (B)
- *is_sponsored* – Whether the product is a Sponsored (paid promotion) item or Organic (no paid promotion) (B)
- *is_couponed* – Special discounted coupons availability (B)
- *Buy_box* – BuyBox button availability on amazon search page, like ‘add to cart’ (B)

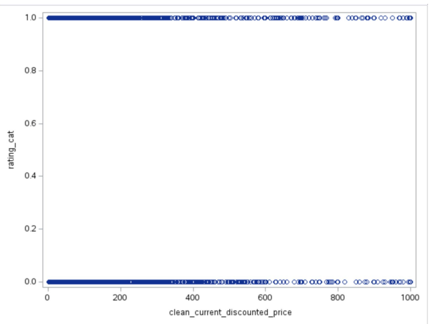
Y variable

- *rating_cat* – Whether product rating is greater than or equal to 4.5/5 stars or not (B)

Complete Separation: Pre-Analysis

We visualized scatter plots for our quantitative variables (*clean_number_reviews* and *clean_current_discount_price*) and found overlap between the points for each possible value of y (low vs. high rating), indicating no complete separation.

Then, we made frequency tables for all categorical variables (*is_discount*, *is_best_seller_clean*, *is_sponsored*, *is_couponed_clean* and *buy_box_clean*) by rating_cat, and found no 'Percents' of 0. Again, indicating no complete separation.



The FREQ Procedure

Frequency
Percent
Row Pct
Col Pct

Table of is_discount by rating_cat

is_discount	rating_cat: 0	rating_cat: 1	Total
Discount	5108 17.44 43.65 40.73	6593 22.51 56.35 39.36	11701 39.95
No Discount	7432 25.37 42.26 59.27	10156 34.68 57.74 60.64	17588 60.05
Total	12540 42.81	16749 57.19	29289 100.00

Why GLM - Logistic Regression

rating_cat only takes on the values of 0 or 1. It is binary or Bernoulli.

- The mean and variance are directly related, and therefore, the distribution is not normal.
- We cannot use OLR as it could give us estimates greater than 1 or less than 0, and it will not be a good representation.

The model will use maximum likelihood estimation (MLE) instead of least squares to estimate, and a logit link function to link the parameter of interest (πi) to the linear predictor to ensure that our predicted value for $Y i$ will have a domain in [0,1].

Fully Specified Model

For i 1,...,29,289, Yi = 1 if the product is 'Highly Rated', or a rating of 4.5/5 stars or above, Yi = 0 otherwise.

- Let clean_number_reviews i = the total number of customer reviews for product i
- Let Discount i = 1 if the listed price for product i is discounted, 0 otherwise.
- Let clean_current_discounted_price i = the current price of product i after any discounts were applied
- Let Bestseller i = 1 if product i is labeled as a 'best seller', 0 otherwise
- Let Organic i = 1 if product i is 'Organic' (not pay-to-promote), and 0 otherwise..
- Let No i (is_couponed_clean) = 1 if product i does not have a special discounted coupon applied, and 0 otherwise.
- Let No i (buy_box_clean) = 1 if product i does not have a marketed 'add to cart' button on the amazon search page, and 0 otherwise.

Model (GLM - Logistic Regression)

Random Component: Yi ~ Bernoulli(πi), where πi = the probability of product i being highly rated.

Systematic Component:

- **Link Function**: ni = log(πi / 1 - πi)
- ni= ß0 + ß1*(clean_number_reviews) + ß2*(is_discount)
  + ß3*(clean_current_discount) + ß4*(is_best_seller_clean)
  + ß5*(is_sponsored) + ß6*(is_couponed_clean)
  + ß7*(buy_box_clean)

Results

Analysis of Maximum Likelihood Estimates						
Parameter		DF	Estimate	Standard Error	Wald Chi-Square	Pr > ChiSq
Intercept		1	-0.9114	0.0584	243.6972	<.0001
clean_number_reviews		1	0.000023	2E-6	130.1078	<.0001
is_discount	Discount	1	-0.2525	0.0256	97.1008	<.0001
clean_current_discou		1	-0.00170	0.000066	669.7834	<.0001
is_best_seller_clean	Best Seller	1	0.2652	0.0774	11.7322	0.0006
is_sponsored	Organic	1	0.1669	0.0307	29.5136	<.0001
is_couponed_clean	No	1	1.5160	0.0612	614.2606	<.0001
buy_box_clean	No	1	-0.7989	0.0411	376.9533	<.0001

ni = -0.9114 + 0.000023*(clean_number_reviews) - 0.2525*(is_discount)
-0.0017*(clean_current_discount) + 0.2652*(is_best_seller_clean)
+0.1669*(is_sponsored) + 1.5160*(is_couponed_clean) - 0.7989*(buy_box_clean)

Complete Separation: Post-Analysis

The standard error for each variable is very small and the p-values are all significant. Therefore, we can confidently say that we do not have complete separation, and can use our model as is, *without* firth.

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is_sponsored	Organic	1	0.1669	0.0307	29.5136	<.0001
is_couponed_clean	No	1	1.5160	0.0612	614.2606	<.0001
buy_box_clean	No	1	-0.7989	0.0411	376.9533	<.0001

Hypothesis Testing: Full Model

Ho: ß1 = ß2 = ß3 = ß4 = ß5 = ß6 = ß7 = 0

Ha: at least 1 ß ≠ 0

Test statistic = lr = 2245.16

X^2(7) , df = 7

P-value = <.0001

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	2245.1572	7	<.0001
Score	2113.3728	7	<.0001
Wald	1918.7123	7	<.0001

Conclusion: We Reject Ho, in favor of Ha. At least 1 ß ≠ 0. *Therefore, we can conclude that our overall model is a significant predictor of rating_cat.*

Hypothesis Testing: Numeric

Testing ß1 (*clean_number_reviews*)

Ho: ß1 = 0

Ha: ß1 ≠ 0

Test statistic = w = 130.11

X^2(1), df = 1

P-value = <.0001

Type 3 Analysis of Effects			
Effect	DF	Wald Chi-Square	Pr > ChiSq
clean_number_reviews	1	130.1078	<.0001
is_discount	1	97.1008	<.0001
clean_current_discou	1	669.7834	<.0001
is_best_seller_clean	1	11.7322	0.0006
is_sponsored	1	29.5136	<.0001
is_couponed_clean	1	614.2606	<.0001
buy_box_clean	1	376.9533	<.0001

Conclusion: We Reject Ho, in favor of Ha. ß1 ≠ 0. *Therefore, we can conclude that clean_number_reviews is a significant predictor of rating_cat.*

Hypothesis Testing: Categorical

Testing ß5 (*is_sponsored*)

Ho: ß5 = 0

Ha: ß5 ≠ 0

Test statistic = w = 29.51

X^2(1), df = 1

P-value = <.0001

Conclusion: We Reject Ho, in favor of Ha. ß5 ≠ 0. *Therefore, we can conclude that is_sponsored is a significant predictor of rating_cat.*

Testing ß6 (*is_couponed_clean*)

Ho: ß6 = 0

Ha: ß6 ≠ 0

Test statistic = w = 614.26

X^2(1), df = 1

P-value = <.0001

Conclusion: We Reject Ho, in favor of Ha. ß6 ≠ 0. *Therefore, we can conclude that is_couponed_clean is a significant predictor of rating_cat.*

Continuing Analysis

We found that not only was the model as a whole significant, but **all** of the explanatory variables within our model are significant predictors of rating. We decided to focus on **sponsorship, coupons, and number of reviews** for our analysis.

Baseline Reference Assumptions

- **No** Discount
- **Not** labeled as a ‘Best Seller’
- Paid Sponsorship (not Organic)
- **With** a coupon
- **With** a button such as “Add to cart”

Analyzing Product Sponsorship

The model **estimates that odds** of a product that is organically promoted obtaining a rating of 4.5 stars or above **change by a factor of 1.18** compared to products with general sponsored products, all else held constant.

That means organically sponsored items are 18% times more likely to be higher rated than pay-to-promote items.

We are 95% confident that **the odds** of a highly-reviewed item **change by a factor between 1.11 and 1.26** for an organically sponsored item compared to a pay-to-promote item.

Therefore, we are 95% confident that the organically sponsored items are 11%-26% more likely to be higher rated than pay-to-promote items.

Estimated Odds Formula: ß5 estimate = 0.1669, e^(ß5 estimate) = 1.18

Confidence Interval Formula: e^(lowerß5 = 0.1066) and e^(upperß5 =0.2271) = (1.11,1.26)

Parameter Estimates and Profile-Likelihood Confidence Intervals				
Parameter		Estimate	95% Confidence Limits	
is_sponsored	Organic	0.1669	0.1066	0.2271

Analyzing the Use of Coupons

The model **estimates that odds** of a non-couponed product obtaining a rating of 4.5 stars or above **change by a factor of 4.55** compared to products with a coupon, all else held constant.

That means non-couponed items are 4.55 times more likely to be higher rated than coupon items.

We are 95% confident that the **odds** of a highly-reviewed item **change by a factor between 4.04 and 5.14** for a non-couponed item compared to a coupon item.

Therefore, we are 95% confident that the non-couponed items are 4-5 times more likely to be higher rated than coupon items.

Estimated Odds Formula: ß6 estimate = 1.5160, e^(ß6 estimate)

Confidence Interval Formula: e^(lowerß1 = 1.397) and e^(upperß1 =1.6368)

Parameter Estimates and Profile-Likelihood Confidence Intervals

Parameter		Estimate	95% Confidence Limits	
is_couponed_clean	No	1.5160	1.3970	1.6368

Analyzing the Number of Reviews

For every additional review added, the **estimated odds** of obtaining a rating of 4.5 stars or above **change by a factor of 1.000023**, all else held constant.

That means our model predicts that the estimated odds of a high rating increase by 0.0023% for each review added.

While this is significant, it is a very small percentage of increase. This makes sense because ß1 is very small. *If we change the number of reviews to a higher value instead of 1…*

Estimated Odds Formula: ß1 estimate = 0.000023, = e^(ß1 estimate)

Parameter Estimates and Profile-Likelihood Confidence Intervals				
Parameter		Estimate	95% Confidence Limits	
clean_number_reviews		0.000023	0.000019	0.000027

Analyzing the Number of Reviews

For every **1,000** increase in number of reviews the **estimated odds** of obtaining a rating of 4.5 stars or above **change by a factor of 1.023**, while they **change by a factor of 1.2586 for every 10,000** increase, all else held constant.

That means our model predicts that the estimated odds of a high rating only increase by 2.3% for each 1,000 reviews added, but increase by 25.86% for each 10,000 reviews added.

We are 95% confident that the addition of 1,000 reviews **changes the odds** of a 4.5 star or higher rating **by a factor between 1.019 and 1.027**, and the addition of 10,000 reviews **changes the odds by a factor between 1.2092 and 1.3099**, holding all else constant.

Therefore, we are 95% confident that the true odds of a high rating are between 1.9% and 2.7% for every 1,000 reviews added, while the true odds are between 21% and 31%, for every 10,000 reviews added.

Custom Odds Ratio Formula (10,000): ß1 estimate = 0.000023, (Odds at x+10,000)/(Odds at x) = e^(10,000*ß1 estimate), **Confidence Interval Formula:** e^(10,000*lowerß1 = 0.000019) and e^(10,000*upperß1 = 0.000027)

Custom Odds Ratio Formula (1,000): ß1 estimate = 0.000023, (Odds at x+1,000)/(Odds at x) = e^(1,000*ß1 estimate), **Confidence Interval Formula:** e^(1,000*lowerß1 = 0.000019) and e^(1,000*upperß1 = 0.000027)

Probability Predictions

A $20 product with 30,000 reviews, no discount, not labeled as a best-seller, with a buy-box button....

1. **without** a coupon, **is not** sponsored (organic) has a **80.65%** probability of being rated 4.5 stars or above.
2. **with** a coupon, **is** sponsored has a **43.65%** probability of being rated 4.5 stars or above.
3. **without** a coupon, **is** sponsored has a **77.91%** probability of being rated 4.5 stars or above.

1. $$\pi = \frac{e^{-0.9114+0.000023(\underline{30000})-0.2525(0)-0.00170(\underline{20})+0.2652(0)+0.1669(\underline{1})+1.5160(\underline{1})-0.7989(0)}}{1+e^{-0.9114+0.000023(\underline{30000})-0.2525(0)-0.00170(\underline{20})+0.2652(0)+0.1669(\underline{1})+1.5160(\underline{1})-0.7989(0)}} = 0.806511$$

2. $$\pi = \frac{e^{-0.9114+0.000023(\underline{30000})-0.2525(0)-0.00170(\underline{20})+0.2652(0)+0.1669(0)+1.5160(0)-0.7989(0)}}{1+e^{-0.9114+0.000023(\underline{30000})-0.2525(0)-0.00170(\underline{20})+0.2652(0)+0.1669(0)+1.5160(0)-0.7989(0)}} = 0.436450$$

3. $$\pi = \frac{e^{-0.9114+0.000023(\underline{30000})-0.2525(0)-0.00170(\underline{20})+0.2652(0)+0.1669(0)+1.5160(\underline{1})-0.7989(0)}}{1+e^{-0.9114+0.000023(\underline{30000})-0.2525(0)-0.00170(\underline{20})+0.2652(0)+0.1669(0)+1.5160(\underline{1})-0.7989(0)}} = 0.779129$$

90% Probability of High-Rating

We estimate that there is a **90% chance** that a $20 product will be rated 4.5+ stars if it has no discount, is not labeled as a best-seller, has a buy-box button....

1. **not** sponsored (organic), **does not** have a coupon, **AND it has 63,467 reviews.**
2. **is** sponsored, **does** have a coupon **AND it has 136,636 reviews.**
3. **is** sponsored, **does not** have a coupon, **AND it has 70,723 reviews.**

Formula 1: log(0.9/1-0.9) = -0.9114 + 0.000023*(clean_number_reviews = X) - 0.2525*(is_discount = 0) -0.0017*(clean_current_discount = 20) + 0.2652*(is_best_seller_clean = 0) +0.1669*(is_sponsored = 1) + 1.5160*(is_couponed_clean = 1) - 0.7989*(buy_box_clean = 0) = 63,466.29

Formula 2: log(0.9/1-0.9) = -0.9114 + 0.000023*(clean_number_reviews = X) - 0.2525*(is_discount = 0) -0.0017*(clean_current_discount = 20) + 0.2652*(is_best_seller_clean = 0) +0.1669*(is_sponsored = 0) + 1.5160*(is_couponed_clean = 0) - 0.7989*(buy_box_clean = 0) = 136,635.85

Formula 3: log(0.9/1-0.9) = -0.9114 + 0.000023*(clean_number_reviews = X) - 0.2525*(is_discount = 0) -0.0017*(clean_current_discount = 20) + 0.2652*(is_best_seller_clean = 0) +0.1669*(is_sponsored = 0) + 1.5160*(is_couponed_clean = 1)) - 0.7989*(buy_box_clean = 0) = 70,722.81

Recommendations to Client

Organically sponsored items without coupons.

- We are 95% confident that the non-couponed items are **4-5 times more likely** to be higher rated than coupon items, holding all else constant.
- We are 95% confident that the organically sponsored items are **11%-26% more likely** to be higher rated than pay-to-promote items holding all else constant.
- With 30,000 reviews, Organically sponsored items without coupons had **the highest probability (80.65%)** of being highly-rated
- Organically sponsored items without coupons **required the least amount of reviews (63,467)** to reach a 90% probability of high-rating.

The more reviews, the better chance of a higher rating.

- We are 95% confident that the odds of a high rating **increase by between 21% and 31%** for each **10,000** reviews added. But, adding a lesser value like **1000** reviews will only increase the odds by ***1.9% - 2.7%***. Reviews become more powerful as the amount increases, but this can be difficult to implement because the user is the one reviewing, not Amazon employees.

The more that ratings can be dependent on steps that Amazon takes (sponsorship, couponing), rather than ones more difficult to control (number of reviews), the less predictability and greater impact.

Thank You